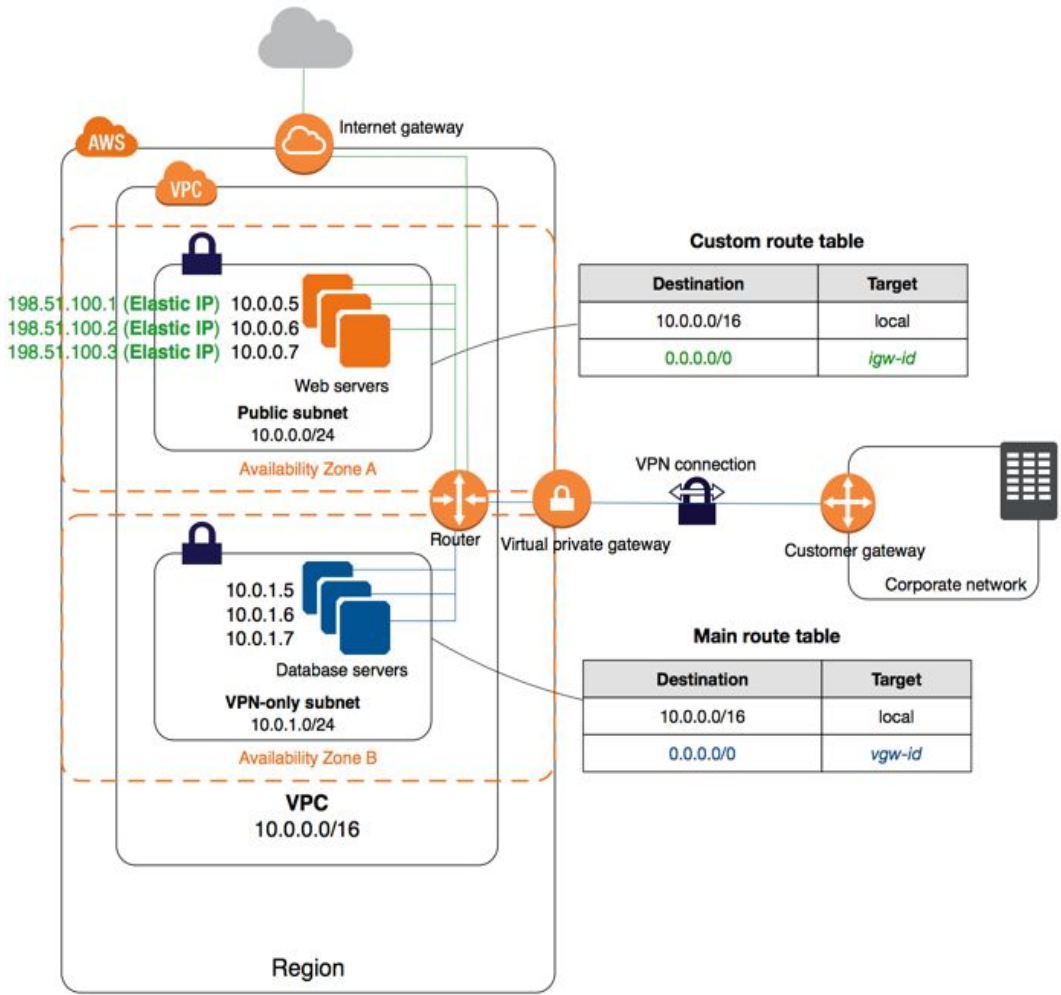


Networking Concept

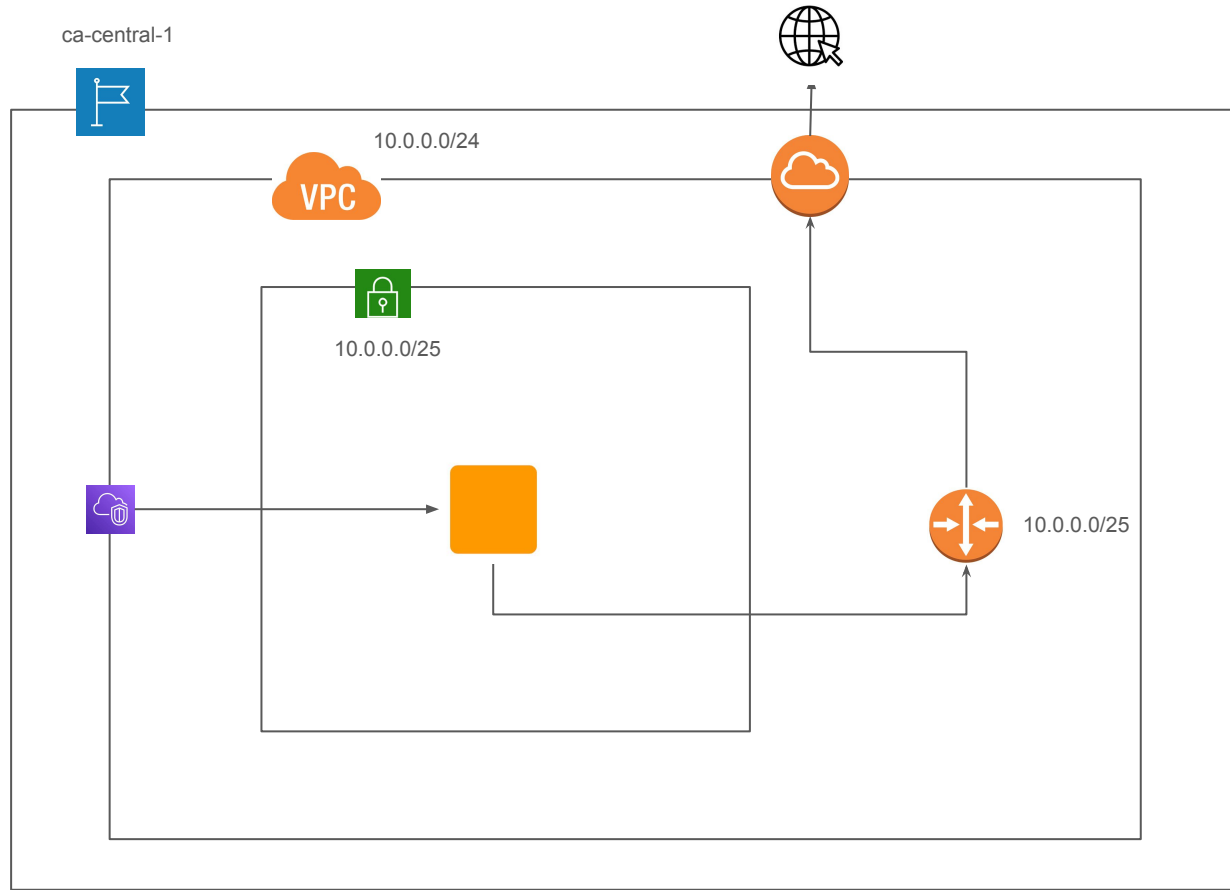
VPC Basics and Cloud practices

What is Amazon VPC?

- Amazon VPC can be referred to as the private cloud inside the cloud.
- It is a logical grouping of servers in a specified network.
- The servers that you are going to deploy in the Virtual Private Cloud(VPC) will be completely isolated from the other servers that are deployed in the Amazon Web Services.
- You can have complete control of the IP address to the virtual machines and route tables and gateways to the VPC.
- With the help of security groups and network access control lists, you can protect your application more.
- Resources and applications are accessed through IPv4 or IPv6 in your AWS VPC.
- It gives the benefit of scalable infrastructure in the AWS environment.
- It gives you complete control over your virtual network.



ca-central-1



VPC vs Private Cloud

AWS VPC is a public cloud service that provides a private cloud-like experience, whereas a private cloud is a dedicated cloud environment hosted on-premises or in a colocation facility.

Feature	AWS VPC	Private Cloud
Ownership	Managed by AWS	Managed by the organization
Location	Hosted in AWS data centres	Hosted on-premises or in a colocation facility
Scalability	Highly scalable	Scalable to meet the needs of the organization
Security	Security is managed by AWS	Security is managed by the organization
Cost	Pay-as-you-go pricing	Capital expense and operational expense

Types of AWS VPCs in AWS Cloud

Default VPC	Non Default VPC
virtual network that is automatically created for the customer's AWS account when EC2 resources are provisioned for the first time	Also known as Customer VPC is not created automatically when EC2 resources are provisioned, and the customer must create their own VPC.

- The AWS system automatically creates the default VPC, whereas the customer/non-default VPC must be manually configured by each customer and resources must be provisioned.
- When a new instance is launched without first allocating a subnet, the Default VPC is assigned.
- Another significant advantage of **Default VPC** is that it includes Internet access by default, as well as an internet gateway and public subnets with corresponding route tables.
- This feature is not enabled by default in non-default VPC.
- In fact, in **non-default VPCs**, public IPv4 addresses are not assigned.
- In terms of numbers, only VPC is available per region, whereas customer VPC is limited to 5 by default for each region.

Benefits Of Using AWS Virtual Private Cloud

- EC2 Instance security group membership can be changed while it is running.
- Static IPv4 is assigned to Instances that persist across the start and stop.
- Create a layered network of resources.
- A single-tenant hardware option is available to run EC2 Instances.
- Access Control List (ACL) is an additional security layer to protect Instances.
- Multiple IPv4 can be assigned to your Instances.
- Control both inbound and outbound traffic of Instances.
- Multiple network interfaces can be attached to EC2 Instances.

Components of AWS VPC (Will be discussed in detail post pricing of VPC)

Route Table: In AWS Virtual Private Cloud, route Tables are the set of rules, that are used to determine where the network traffic has to be directed. The route table specifies the destination (IP address) and target (where do want to send the traffic to that destination). The target can be an Internet gateway, NAT gateway, Virtual private gateway, VPC peering connection, etc

Subnet: It is a portion of the network that shares a common address component. All devices whose addresses have the same prefix are in the same subnet. For example, all those devices whose IP address would start with 172.31.1 would be part of the same subnet. There are two types of subnets. **Private Subnet** where resources are not exposed to the outside world and **Public Subnet** where resources are exposed to the internet through Internet Gateway.

Security Groups: Security groups are a set of firewall rules that controls the traffic for your instance. In Amazon Firewall the only action that can be carried out is allowed. You cannot create a rule to deny. The destination is always the instance on which the service security group is running. You can have a single security group associated with multiple instances.

NAT Gateway: Network Address Translation (NAT) Gateway is used when higher bandwidth and availability with lesser administrative effort is required. NAT gateway always resides inside the public subnet of an Availability Zone.

It updates the routing table of the private subnet such that it sends the traffic to the NAT gateway. Elastic IP must be attached to the NAT gateway while creating. It supports only TCP, UDP, and ICMP protocols.

VPC Peering: A VPC peering connection allows you to route traffic between two Virtual Private Cloud's using IPv4 or IPv6 private addresses. Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. A VPC peering connection helps you to facilitate the transfer of data

Network Access Control Lists (NACL): an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets. You might set up network ACLs with rules similar to your security groups in order to add an additional layer of security to your VPC. The default network ACL is configured to allow all traffic to flow in and out of the subnets to which it is associated.

Virtual Private Gateway: A virtual private gateway is the VPN concentrator on the Amazon side of the VPN connection. You create a virtual private gateway and attach it to the VPC from which you want to create the VPN connection.

Customer Gateway: An Amazon VPC VPN connection links your data center (or network) to your Amazon VPC (virtual private cloud). A customer gateway is an anchor on your side of that connection. It can be a physical or software appliance.

Elastic IP: It is a static IP address that never changes and is a reserved public IP address that can be assigned to any Instance in a particular region. An elastic IP is reserved for your AWS account and is yours until you release it.

Network Interface: Network Interface is a point of connection between a public and a private network. Every instance has a default network interface, called the primary network interface. Network traffic is automatically shifted to the new instance if you move it from one instance to the other.

VPC Endpoints: VPC endpoints allow private connection between your AWS VPC and other AWS services without using the internet. VPC endpoint devices are scaled, redundant, and highly available VPC components. There are two types of AWS Virtual Private Cloud endpoints **Interface endpoints** and **Gateway Endpoints**.

Best Practices For Securing Your AWS VPC Implementation

Running a machine with mission-critical workloads requires multiple layers of security. Amazon Virtual Private Cloud can be secured like your on-premises data center by following some of these useful tips:

- Amazon Web Services marketplace offers you a web application firewall, a firewall virtual appliance, and a few other tools which you can use to secure your Amazon VPC.
- To secure your protocols from unauthorized access you can configure intrusion detection systems and intrusion prevention virtual appliances.
- With the help of Configure Privileged Identity access management, you can audit and monitor Administrator access to your VPC.
- For transferring information securely between Amazon VPC among diverse regions or Amazon VPC to an on-premises data center, you can easily configure a Site-to-Site VPN.
- Another option to transfer information securely is to use AWS Transfer for Secure File Transfer Protocol (AWS SFTP). With AWS SFTP, you use VPC endpoints and avoid using public IP addresses or going through the internet. In addition, VPC endpoints for AWS SFTP leverage security functionality via AWS private link, which provides private connections between your VPCs and AWS services.

AWS VPC Pricing

Amazon VPC Traffic Mirroring Pricing

If you choose to enable traffic mirroring on the Elastic Network Interface (ENI) of Amazon EC2 instances, you will be charged hourly for each ENI that is enabled with traffic mirroring. If you no longer wish to be charged for traffic mirroring, simply disable traffic mirroring on EC2 instance ENIs using the AWS Management Console, command-line interface, or API. The hourly price per ENI is: \$0.015

NAT Gateway Pricing

- **NAT Gateway Hourly Charge:** NAT Gateway is charged on an hourly basis. For this region, the rate is \$0.045 per hour.
- **NAT Gateway Data Processing Charge:** 1 GB of data went through the NAT gateway. The NAT Gateway Data Processing charge is applied and will result in a charge of \$0.045.
- **Data Transfer Charge:** This is the standard EC2 Data Transfer charge. 1 GB of data was transferred from the EC2 instance to S3 via the NAT gateway. There was no charge for the data transfer from the EC2 instance to S3 as it is Data Transfer Out to Amazon EC2 to S3 in the same region.

AWS VPN pricing

If you create an AWS Site-to-Site VPN connection to your Amazon VPC, you are charged for each VPN connection hour i.e. \$0.05 per Site-to-Site VPN connection per hour

AWS provides a number of efficient, secure connectivity options to help you get the most out of AWS when integrating your remote networks with Amazon VPC. AWS Virtual Private Cloud enables you to launch AWS resources into a virtual network that you've defined. This virtual network closely resembles a traditional network that you would operate in your own data center, with the benefits of using the scalable infrastructure of AWS.

Amazon VPC (Virtual Private Cloud) Fundamentals

- If the subnet has internet access then it is called Public Subnet.
- If the subnet doesn't have internet access then it is called Private Subnet.
- A subnet must reside entirely within one Availability Zone.
- An entire subnet must be contained within a single Availability Zone.
- Access between instances is managed by VPC Security Groups for both inbound and outgoing traffic (EC2 Security Groups can only define inbound rules).
- We can specify Subnet IP Routing with the aid of the Route Table.
- If a server/instance which is in a private subnet wants to reach the internet then it must have NAT in a public subnet.

Subnet

- **A subnet is a smaller portion of the network that typically includes all the machines in a certain area.**
- **We can add as many as subnets we need in one availability zone. Each subnet must reside entirely within one availability zone.**
- **The public subnets will be attached to Internet Gateway which enables Internet access.**
- **The private subnets will not have internet access.**
- **Each and every subnet which is presented in VPC must be associated with the routing table.**

Internet Gateway

- **With the help of IGW (Internet Gateway), the resources present (e.g: EC2) in the VPC will enable to access the Internet.**
- **One VPC can't have more than one IGW**
- **If resources are running in a certain VPC then IGW can not be detached from that particular VPC.**

Route Table

- **Route Table contains a set of rules, called route which helps us to route the network traffic.**
- **A single VPC can have as many as route tables it requires.**
- **If the dependencies are attached to the route table then they can't be deleted.**

NACL Network Access Control Lists

- **The NACL security layer for VPC serves as a firewall to manage traffic entering and leaving one or more subnets.**
- **The NACL for the default VPC is active and connected to the default subnets.**

Classless Inter-Domain Routing (CIDR)

- **A technique for allocating IP addresses and for IP routing is called classless Inter-Domain Routing (CIDR), and its range is 0-32.**
- **When setting up a VPC, we must specify a set of IPv4 addresses using classless Inter-Domain Routing (CIDR),
for (Example:10.0.0.0/16 For our VPC, this will serve as the main CIDR block).**

RFC1918 Address(Private address)

- An enterprise organization will give an internal host an IP address known as an RFC1918 address. These IP addresses are employed in private networks that cannot be accessed or accessed through the internet.
- The following networks are included in the RFC1918 address(Private address)

10.0.0.0 – 10.255.255.255 (10/8 prefix)

172.16.0.0 – 172.31.255.255 (172.16/12 prefix)

192.168.0.0 – 192.168.255.255 (192.168/16 prefix)

What is an IP Address?

All the computers of the world on the Internet network communicate with each other with underground or underwater cables or wirelessly.

If I want to download a file from the internet or load a web page or literally do anything related to the internet, my computer must have an address so that other computers can find and locate mine in order to deliver that particular file or webpage that I am requesting.

In technical terms, that address is called **IP Address or Internet Protocol Address**.

Types of IP Address

1. IPv4: Internet Protocol version 4.

It consists of 4 numbers separated by the dots. Each number can be from 0-255 in decimal numbers.

But computers do not understand decimal numbers, they instead change them to binary numbers which are only 0 and 1.

Therefore, in binary, this (0-255) range can be written as (00000000 – 11111111). Since each number N can be represented by a group of 8-digit binary digits.

So, a whole IPv4 binary address can be represented by 32-bits of binary digits.

In IPv4, a unique sequence of bits is assigned to a computer, so a total of $2^{\text{power } 32}$ devices approximately = 4,294,967,296 can be assigned with IPv4.

Classes of IPv4 Address

- There are around 4.3 billion IPv4 addresses and managing all those addresses without any scheme is next to impossible.
- Let's understand it with a simple example:
- If you have to find a word from a language dictionary, how long will it take? Usually, you will take less than 5 minutes to find that word.
- You are able to do this because words in the dictionary are organized in alphabetical order.
- If you have to find out the same word from a dictionary that doesn't use any sequence or order to organize the words, it will take an eternity to find the word.
- If a dictionary with one billion words without order can be so disastrous, then you can imagine the pain behind finding an address from 4.3 billion addresses.
- For easier management and assignment IP addresses are organized in numeric order and divided into the following 5 classes :

IP Class	Address Range	Maximum number of networks
Class A	1-126	126 (2^7-2)
Class B	128-191	16384
Class C	192-223	2097152
Class D	224-239	Reserve for multitasking
Class E	240-254	Reserved for Research and development

Explanation of IPv4

138.0.0.0 = **10001010.00000000.00000000.00000000**

8bits.8bits.8bits.8bits = 32 bits

1	0	0	0	1	0	1	0
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

192.168.0.0 – 192.168.255.255

$256 \times 256 = 65536$

$(128 \times 2) \times (128 \times 2) = (64 \times 2 \times 2) \times (64 \times 2 \times 2) = (32 \times 2 \times 2 \times 2) \times (32 \times 2 \times 2 \times 2)$

$(16 \times 2 \times 2 \times 2 \times 2) \times (16 \times 2 \times 2 \times 2 \times 2) = (8 \times 2 \times 2 \times 2 \times 2 \times 2) \times (8 \times 2 \times 2 \times 2 \times 2 \times 2)$

$(4 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2) \times (4 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2)$

$(2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2) \times (2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2)$

$= (2^8) \times (2^8)$

$= 2^{(8+8)}$

$= 2^{16}$

$\rightarrow /16 = 192.168.0.0 \rightarrow 0 - 32 \rightarrow 32 - 16 = 16$

2^{16}

/28

$$32 - 28 = 4$$

$$2^4 = 16$$

8 per subnet

RFC1918 - 2 IPs

AWS blocks - 3 IPs

5 IPs blocked per subnet

$$\text{IPs allocated per subnet} = 8 - 5 = 3$$

/25

Ask is to have 2 subnets per az

Total subnets 4

$$/25 \rightarrow 32 - 25 = 7 = 2^7 = 128$$

$$128/4 = 32$$

$$2^5 = 32$$

$$32 - 5 = 27$$

17.16.15.0/24

17.16.15.0 - 17.16.15.255

0 to 255 = 256

$2^8 \rightarrow 32 - 8 = 24$

17.16.15.32 - 17.16.15.255

32 to 255 = 128

256 128 64 32 16 8 4 2 1

PartA.PartB.PartC.PartD

8bits.8bits.8bits.8bits = 32 bits

0 0 0 0 0 0 0 0

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

$128+64+32+16+8+4+2+1= 255$

172.31.0.0/16 = 10101100.00011111 .00000000.00000000

Netmask: 255.255.0.0 = 16 11111111.11111111 .00000000.00000000

Wildcard: 0.0.255.255 00000000.00000000 .11111111.11111111

=>

Network: 172.31.0.0/16 10101100.00011111 .00000000.00000000
(Class B)

Broadcast: 172.31.255.255 10101100.00011111 .11111111.11111111

HostMin: 172.31.0.1 10101100.00011111 .00000000.00000001

HostMax: 172.31.255.254 10101100.00011111 .11111111.11111110

Hosts/Net: 65534 (Private Internet)

VPC $\rightarrow$ 172.31.0.0/16 = 65536

172.31.0.0/20 = 4096

4096 x 6 = 24576

65536 - 24576 = 40960

1Subnet $\rightarrow$ 172.31.0.0/20 $\rightarrow$ 1a

2Subnet $\rightarrow$ 172.31.16.0/20 $\rightarrow$ 1b

3Subnet $\rightarrow$ 172.31.32.0/20 $\rightarrow$ 1c

4Subnet $\rightarrow$ 172.31.48.0/20 $\rightarrow$ 1d

5Subnet $\rightarrow$ 172.31.64.0/20 $\rightarrow$ 1e

6Subnet $\rightarrow$ 172.31.80.0/20 $\rightarrow$ 1f

PartA.PartB.PartC.PartD/16 $\rightarrow$ 0 - 255, 0 - 255

Part C $\rightarrow$ 0,1,2,3,4.....,255 = 256

Part D $\rightarrow$ 0,1,2,3,4.....,255 = 256

256 x 256 = 65536

PartA.PartB.PartC.PartD/8 $\rightarrow$ 0 - 255, 0 - 255, 0 - 255

Part B $\rightarrow$ 0,1,2,3,4.....,255 = 256

Part C $\rightarrow$ 0,1,2,3,4.....,255 = 256

Part D $\rightarrow$ 0,1,2,3,4.....,255 = 256

256 x 256 x 256 = 16777216

VPC → 172.31.0.0/16 = 65536

Subnet → 172.31.0.0/20 = 4096

32-20 = 12

$2^{12} = 4096$

AWS → 5 reserved

$4096 - 5 = 4091 \times 6 = 24546$

172.31.0.0 → 1 IPv4

172.31.0.1 → 2 IPv4

172.31.0.255 → 256 IPv4

172.31.1.0 → 257 IPv4

172.31.15.255 → 4096 IPv4

Understanding CIDR – Subnet Mask

- The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP

192	168	0	0	/32 => allows for 1 IP (2^0)	—————> 192.168.0.0
192	168	0	0	/31 => allows for 2 IP (2^1)	—————> 192.168.0.0 -> 192.168.0.1
192	168	0	0	/30 => allows for 4 IP (2^2)	—————> 192.168.0.0 -> 192.168.0.3
192	168	0	0	/29 => allows for 8 IP (2^3)	—————> 192.168.0.0 -> 192.168.0.7
192	168	0	0	/28 => allows for 16 IP (2^4)	—————> 192.168.0.0 -> 192.168.0.15
192	168	0	0	/27 => allows for 32 IP (2^5)	—————> 192.168.0.0 -> 192.168.0.31
192	168	0	0	/26 => allows for 64 IP (2^6)	—————> 192.168.0.0 -> 192.168.0.63
192	168	0	0	/25 => allows for 128 IP (2^7)	—————> 192.168.0.0 -> 192.168.0.127
192	168	0	0	/24 => allows for 256 IP (2^8)	—————> 192.168.0.0 -> 192.168.0.255
...					
192	168	0	0	/16 => allows for 65,536 IP (2^{16})	—————> 192.168.0.0 -> 192.168.255.255
...					
192	168	0	0	/0 => allows for All IPs	—————> 0.0.0.0 -> 255.255.255.255



Quick Memo

Octets

1st 2nd 3rd 4th

- /32 – no octet can change
- /24 – last octet can change
- /16 – last 2 octets can change
- /8 – last 3 octets can change
- /0 – all octets can change

Public vs. Private IP (IPv4)

- The Internet Assigned Numbers Authority (IANA) established certain blocks of IPv4 addresses for the use of private (LAN) and public (Internet) addresses
- * Private IP can only allow certain values:
 - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) ← in big networks
 - 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) ← AWS default VPC in that range
 - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) ← e.g., home networks
- * All the rest of the IP addresses on the Internet are Public

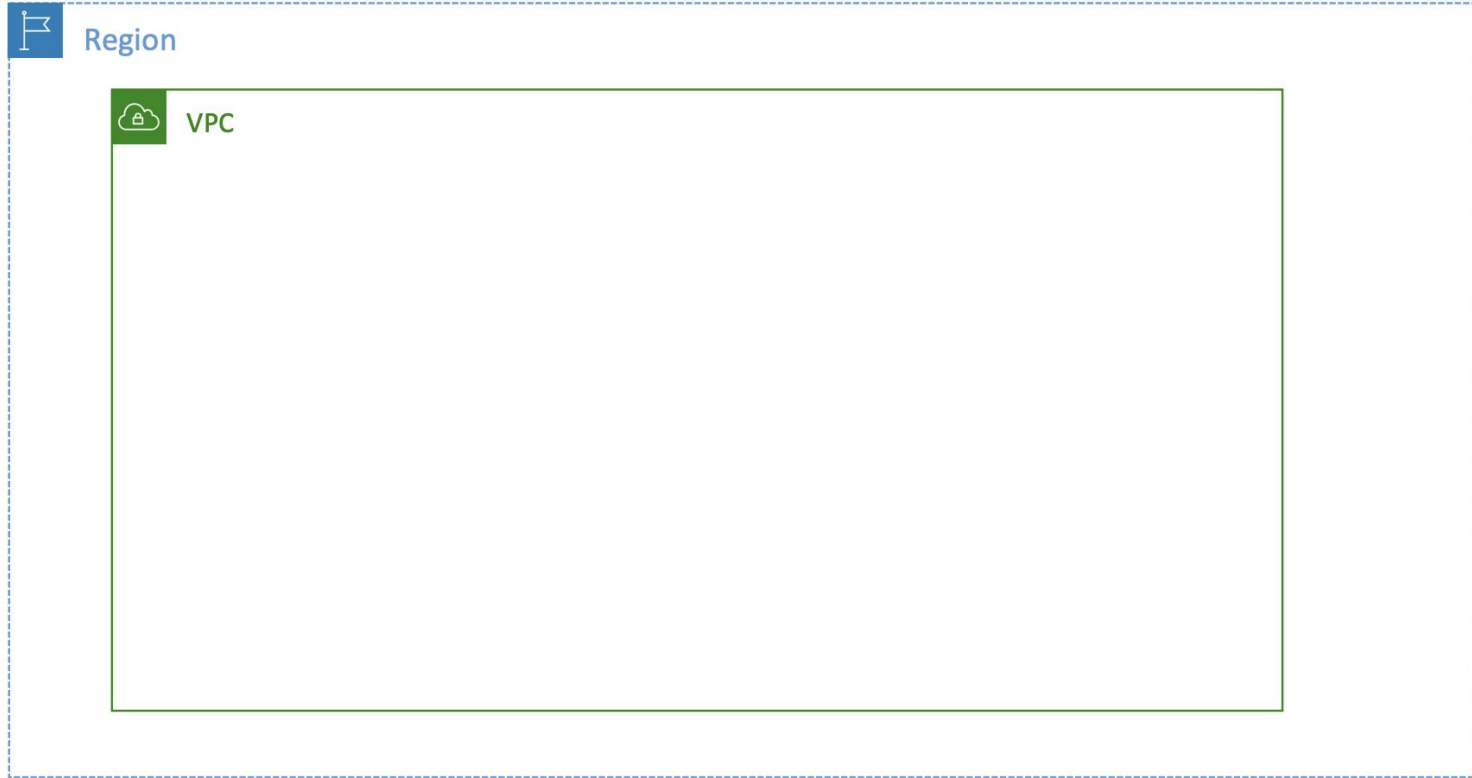
Default VPC Walkthrough

- * All new AWS accounts have a default VPC
 - New EC2 instances are launched into the default VPC if no subnet is specified
 - Default VPC has Internet connectivity and all EC2 instances inside it have public IPv4 addresses
 - We also get a public and a private IPv4 DNS names

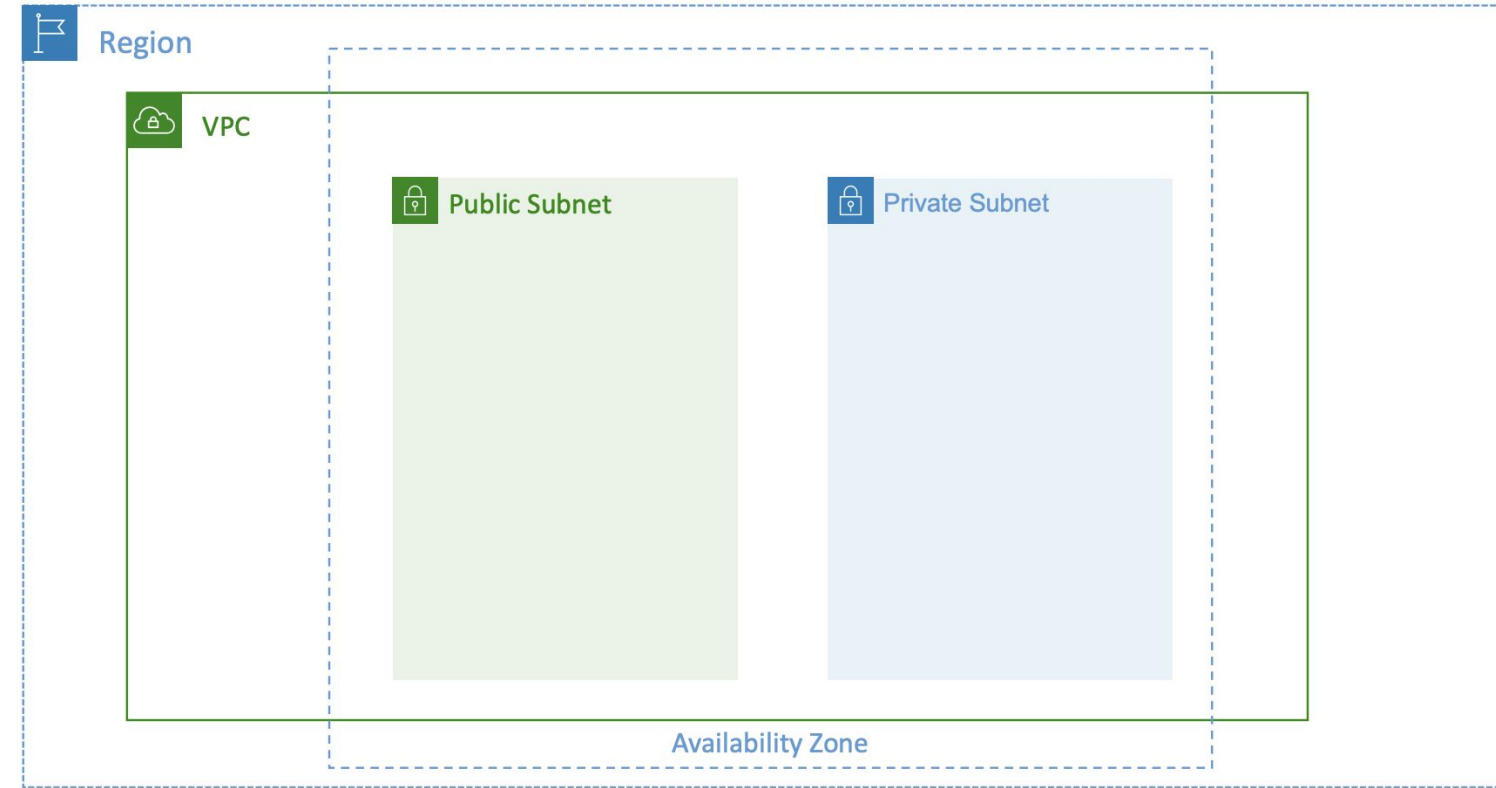
VPC in AWS – IPv4

- VPC = Virtual Private Cloud
- You can have multiple VPCs in an AWS region (max. 5 per region – soft limit)
- Max. CIDR per VPC is 5, for each CIDR:
- Min. size is /28 (16 IP addresses)
- Max. size is /16 (65536 IP addresses)
- Because VPC is private, only the Private IPv4 ranges are allowed:
- 10.0.0.0 – 10.255.255.255 (10.0.0.0/8)
- 172.16.0.0 – 172.31.255.255 (172.16.0.0/12)
- 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)
- Your VPC CIDR should NOT overlap with your other networks (e.g., corporate)

State of Hands -on



Adding Subnets



VPC – Subnet (IPv4)

- AWS reserves 5 IP addresses (first 4 & last 1) in each subnet
- These 5 IP addresses are not available for use and can't be assigned to an EC2 instance
- Example: if CIDR block 10.0.0.0/24, then reserved IP addresses are:
 - 10.0.0.0 – Network Address
 - 10.0.0.1 – reserved by AWS for the VPC router
 - 10.0.0.2 – reserved by AWS for mapping to Amazon-provided DNS
 - 10.0.0.3 – reserved by AWS for future use
 - 10.0.0.255 – Network Broadcast Address.

AWS does not support broadcast in a VPC, therefore the address is reserved

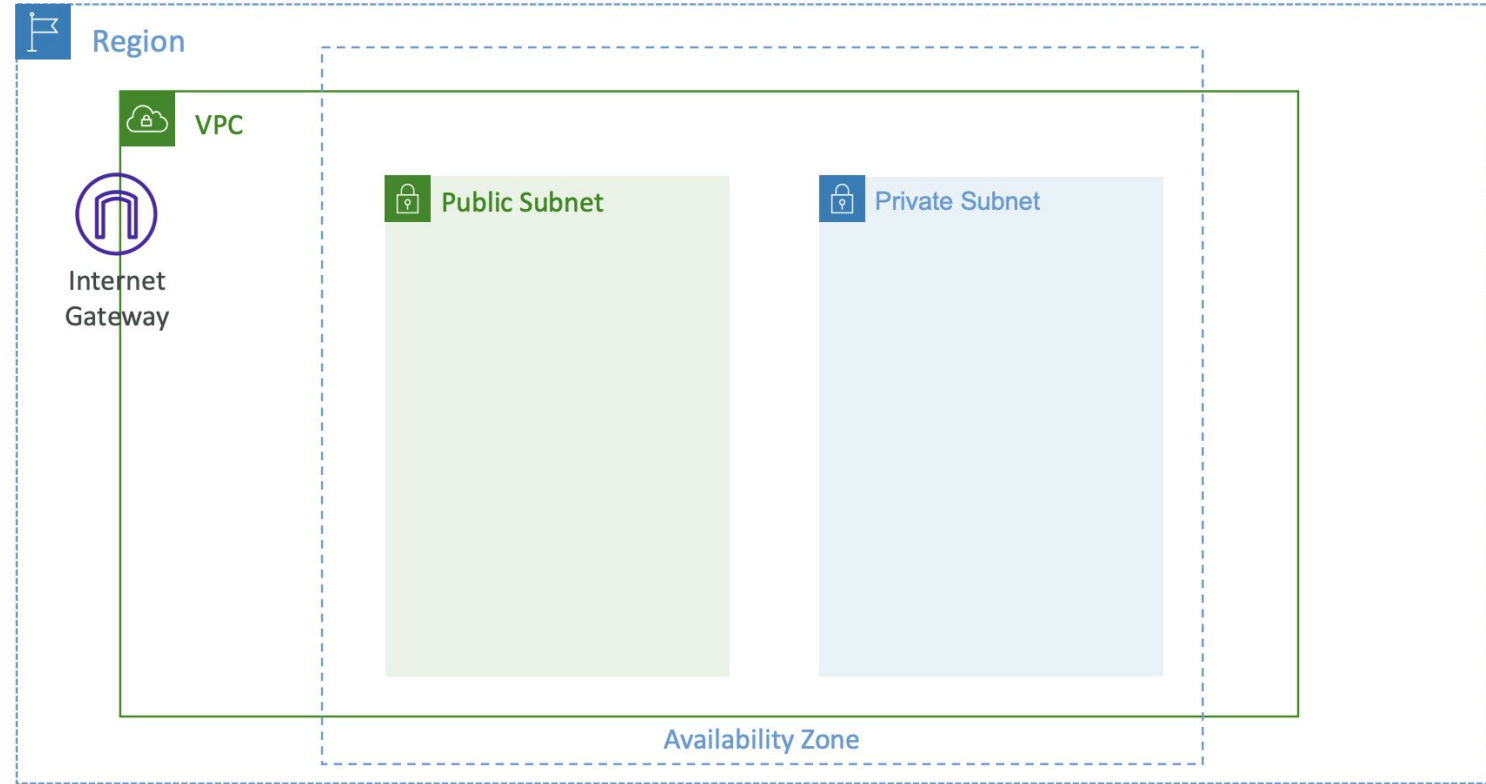
- Exam Tip, if you need 29 IP addresses for EC2 instances:
 - You can't choose a subnet of size /27 (32 IP addresses, $32 - 5 = 27 < 29$)
 - You need to choose a subnet of size /26 (64 IP addresses, $64 - 5 = 59 > 29$)

Internet Gateway (IGW)

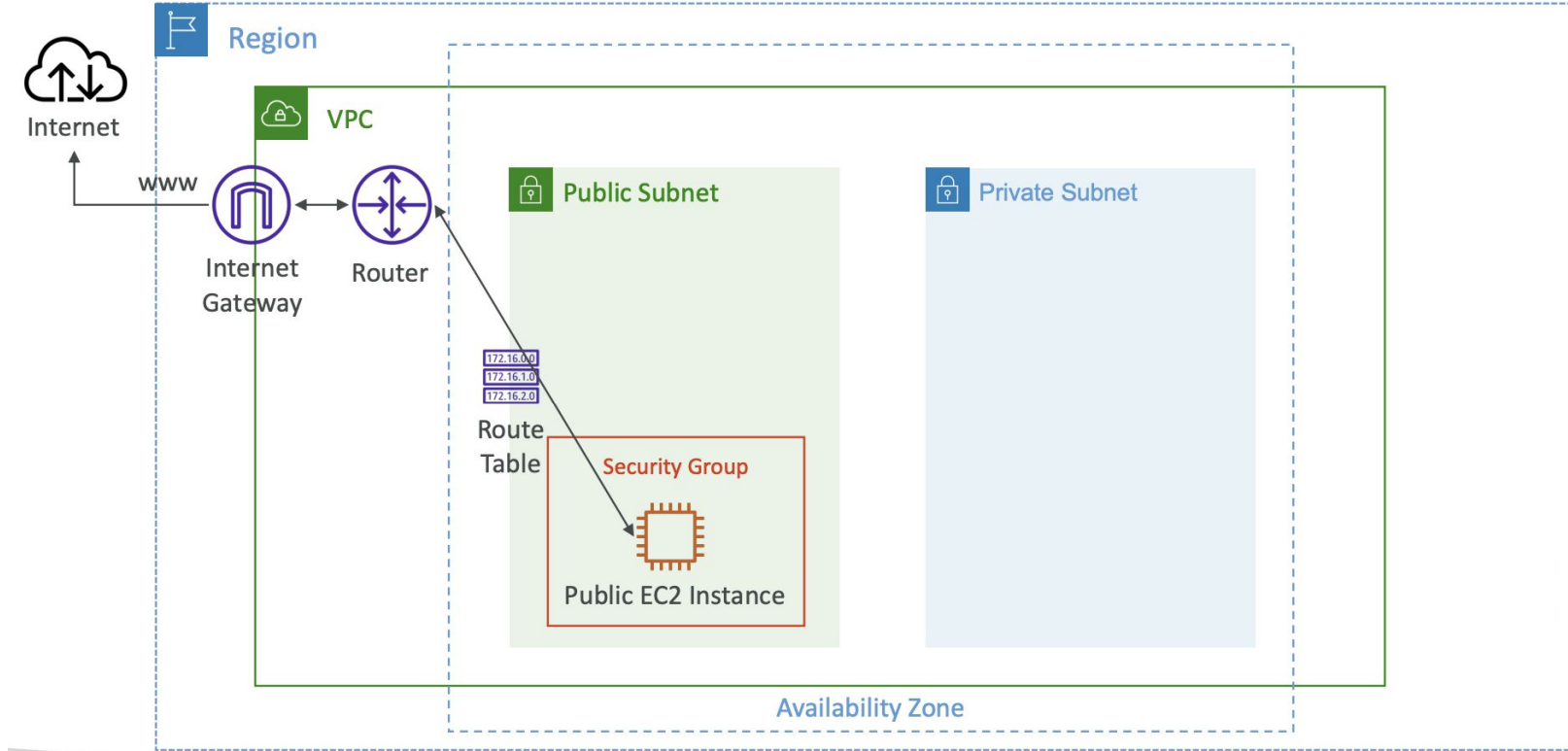
Allows resources (e.g., EC2 instances) in a VPC connect to the Internet

- It scales horizontally and is highly available and redundant
- Must be created separately from a VPC
- One VPC can only be attached to one IGW and vice versa
- Internet Gateways on their own do not allow Internet access...
- Route tables must also be edited!

Adding Internet Gateway



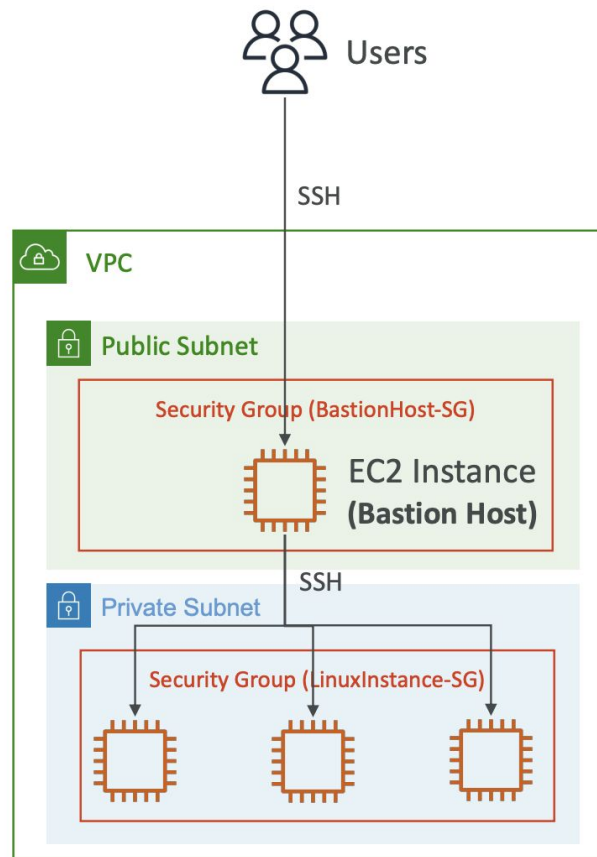
Editing Route Tables



Bastion Hosts

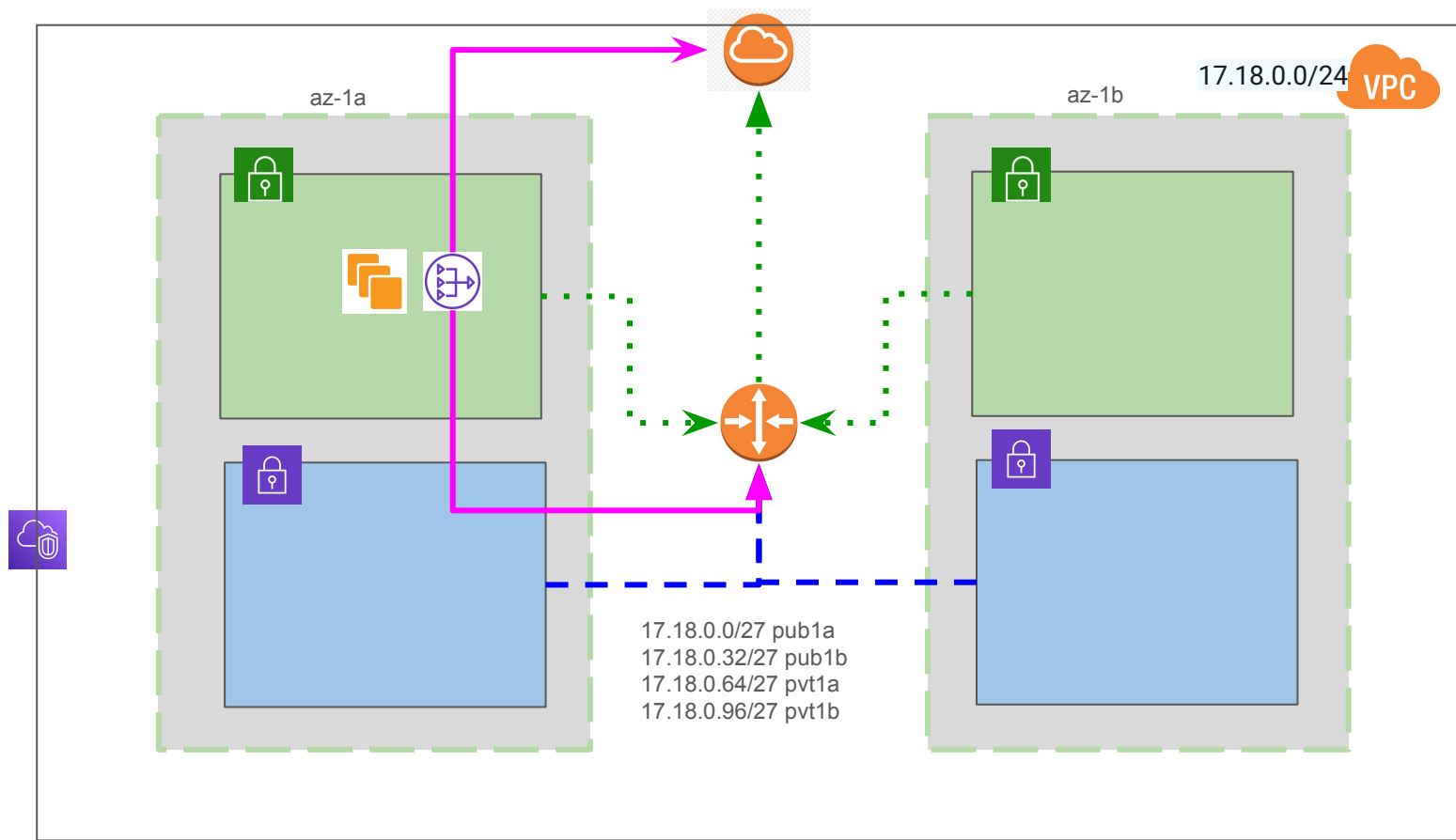
We can use a Bastion Host to SSH into our private EC2 instances.

- The bastion is in the public subnet which is then connected to all other private subnets
- Bastion Host security group must allow inbound from the internet on port 22 from restricted CIDR, for example the public CIDR of your corporation
- Security Group of the EC2 Instances must allow the Security Group of the Bastion Host, or the private IP of the Bastion host





us-east-1



IP discussion

17.18.0.0/27

$32 - 27 = 5$

$2^5 = 32$

32 IPs

$32 - 2 = 30 \rightarrow$ RFC 1918

$30 - 3 = 27 \rightarrow$ AWS

17.18.0.0

17.18.0.1

17.18.0.2

17.18.0.3

17.18.0.4

17.18.0.5

.....

17.18.0.31

Broadcast IP



2048.avsdevops.com



4 NS provide by R53

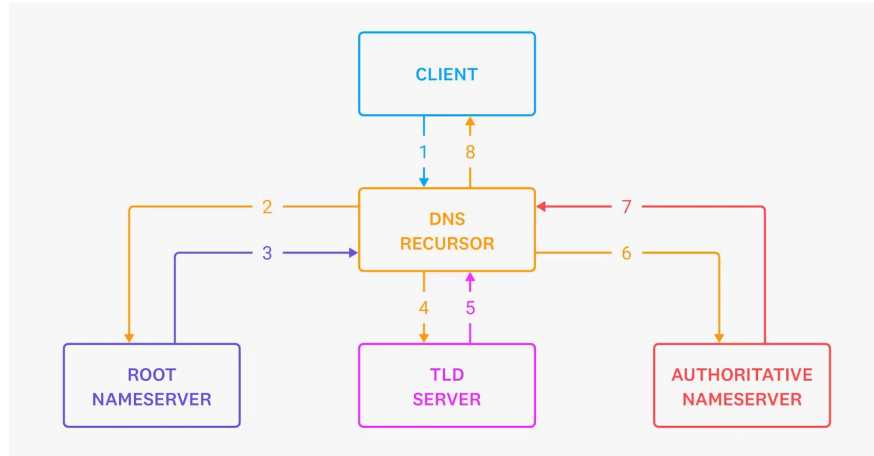


DNS entry A record of
ALB (Layer 7)

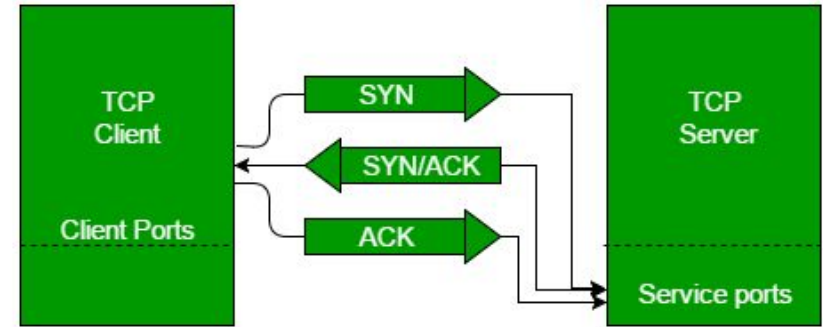


How do we access information from internet?

- DNS resolution



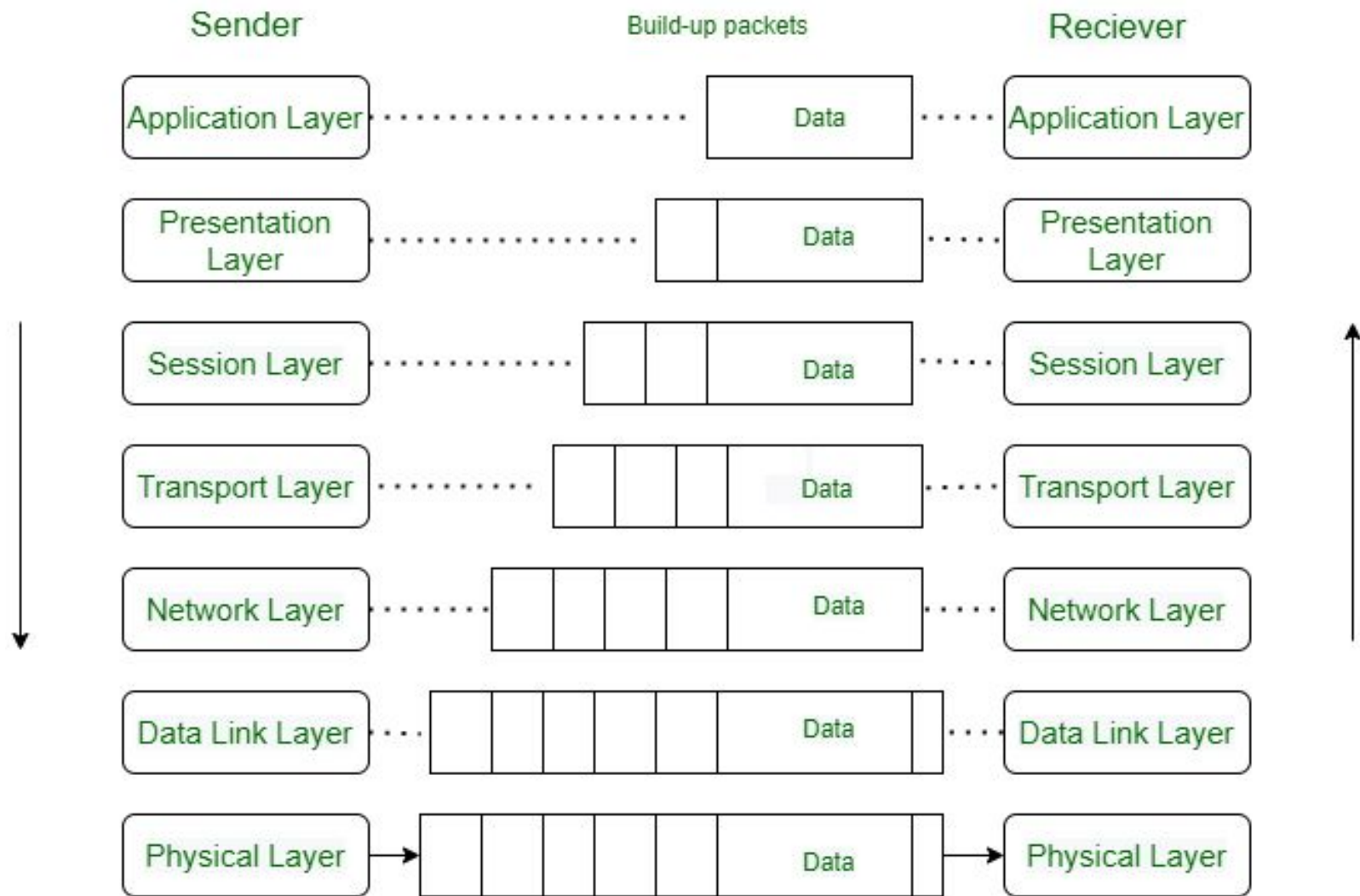
- TCP handshake



What is OSI Model? – Layers of OSI Model

OSI stands for Open Systems Interconnection. It is a 7-layer architecture with each layer having specific functionality to perform. All these 7 layers work collaboratively to transmit the data from one person to another across the globe. OSI model was developed by **ISO** – ‘**International Organization for Standardization**’, in the year 1984.

It is a reference framework that explains the process of transmitting data between computers. It is divided into seven layers that work together to carry out specialised network functions, allowing for a more systematic approach to networking.



OSI MODEL

Data	Application Network Process to Application
Data	Presentation Data Representation and Encryption
Data	Session Interhost Communication
Segments	Transport End-to-end connections and reliability
Packets	Network Path Determination and IP (logical addressing)
Frames	Data Link Physical Addressing
Bits	Physical Media, Signal and Binary Transmission

What are the 7 layers of the OSI Model?

The OSI model consists of seven abstraction layers arranged in a top-down order:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

What is the Flow of Data in OSI Model?

When we transfer information from one device to another, it travels through 7 layers of OSI model. First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end.

Let's look at it with an Example:

Luffy sends an e-mail to his friend Zoro.

Step 1: Luffy interacts with e-mail application like Gmail, outlook, etc. Writes his email to send. (This happens in **Layer 7: Application layer**)

Step 2: Mail application prepares for data transmission like encrypting data and formatting it for transmission. (This happens in **Layer 6: Presentation Layer**)

Step 3: There is a connection established between the sender and receiver on the internet. (This happens in **Layer 5: Session Layer**)

Step 4: Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information. (This happens in **Layer 4: Transport Layer**)

Step 5: Addressing of packets is done in order to find the best route for transfer. (This happens in **Layer 3: Network Layer**)

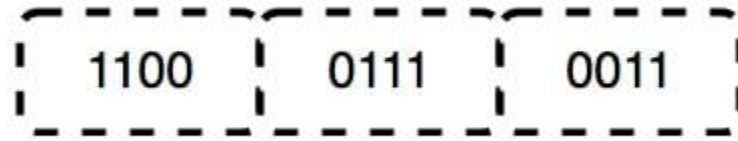
Step 6: Data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection. (This happens in **Layer 2: Data Link Layer**)

Step 7: Lastly Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.

After the email reaches the receiver i.e. Zoro, the process will reverse and decrypt the e-mail content. At last, the email will be shown on Zoro's email client.

Physical Layer – Layer 1

The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of **bits**. It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.



Functions of the Physical Layer

- **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit rate control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.

Data Link Layer (DLL) – Layer 2

The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address. The Data Link Layer is divided into two sublayers:

1. Logical Link Control (LLC)
2. Media Access Control (MAC)

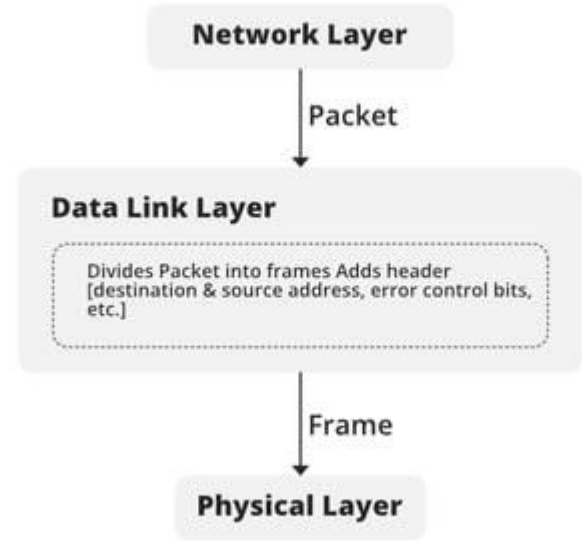
The packet received from the Network layer is further divided into frames depending on the frame size of the NIC(Network Interface Card). DLL also encapsulates Sender and Receiver's MAC address in the header.

The Receiver's MAC address is obtained by placing an ARP(Address Resolution Protocol) request onto the wire asking "Who has that IP address?" and the destination host will reply with its MAC address.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.

- **Physical addressing:** After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.



Note:

1. *Packet in the Data Link layer is referred to as **Frame**.*
2. *Data Link layer is handled by the NIC (Network Interface Card) and device drivers of host machines.*
3. *Switch & Bridge are Data Link Layer devices.*

Network Layer – Layer 3

The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP addresses are placed in the header by the network layer.

Functions of the Network Layer

- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device on Internetwork uniquely, the network layer defines an addressing scheme. The sender & receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.

Note:

1. Segment in the Network layer is referred to as **Packet**.
2. Network layer is implemented by networking devices such as routers and switches.

Transport Layer – Layer 4

The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as *Segments*. It is responsible for the end-to-end delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.

At the sender's side: The transport layer receives the formatted data from the upper layers, performs **Segmentation**, and also implements **Flow and error control** to ensure proper data transmission. It also adds Source and Destination port numbers in its header and forwards the segmented data to the Network Layer.

Note: *The sender needs to know the port number associated with the receiver's application.*

Generally, this destination port number is configured, either by default or manually. For example, when a web application requests a web server, it typically uses port number 80, because this is the default port assigned to web applications. Many applications have default ports assigned.

At the receiver's side: Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.

Services Provided by Transport Layer: *The transport layer is called as **Heart of the OSI** model.*

1. Connection-Oriented Service: It is a three-phase process that includes

- Connection Establishment
- Data Transfer
- Termination/disconnection

In this type of transmission, the receiving device sends an acknowledgment, back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

2. Connectionless service: It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service.

Note:

1. *Data in the Transport Layer is called **Segments**.*
2. *Transport layer is operated by the Operating System. It is a part of the OS and communicates with the Application Layer by making system calls.*
3. **Device or Protocol Use :** TCP, UDP NetBIOS, PPTP

Session Layer – Layer 5

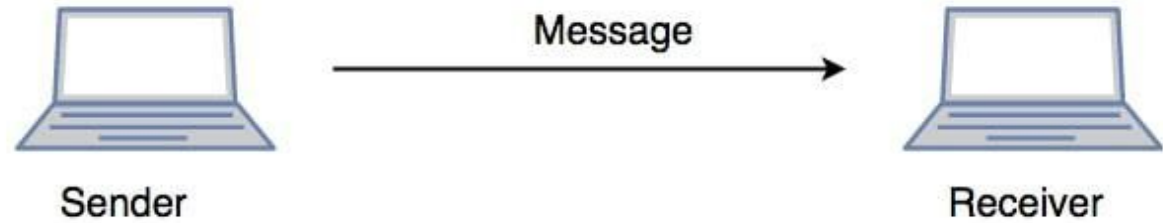
This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.

Functions of the Session Layer

- **Session establishment, maintenance, and termination:** The layer allows the two processes to establish, use, and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

For example:-

Let us consider a scenario where a user wants to send a message through some Messenger application running in his browser. The “Messenger” here acts as the application layer which provides the user with an interface to create the data. This message or so-called Data is compressed, encrypted (if any secure data), and converted into bits (0's and 1's) so that it can be transmitted.



Note:

1. *All the below 3 layers(including Session Layer) are integrated as a single layer in the TCP/IP model as the Application Layer”.*
2. *Implementation of these 3 layers is done by the network application itself. These are also known as **Upper Layers or Software Layers**.*
3. **Device or Protocol Use :** NetBIOS, PPTP.

Presentation Layer – Layer 6

The presentation layer is also called the **Translation layer**. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.

Functions of the Presentation Layer

- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Note: **Device or Protocol Use:** JPEG, MPEG, GIF

Application Layer – Layer 7

At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data, which has to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

Example: Application – Browsers, Skype Messenger, etc.

Note: 1. *The application Layer is also called Desktop Layer.*

2. ***Device or Protocol Use : SMTP***

Functions of the Application Layer

The main functions of the application layer are given below.

- Network Virtual Terminal: It allows a user to log on to a remote host.
- FTAM- File transfer access and management: This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- Mail Services: Provide email service.
- Directory Services: This application provides distributed database sources and access for global information about various objects and services.

Advantages of OSI Model

OSI Model defines the communication of a computing system into 7 different layers. Advantages of OSI Model include:

- It divides network communication into 7 layers which makes it easier to understand and troubleshoot.
- It standardizes network communications, as each layer has fixed functions and protocols.
- Diagnosing network problems is easier with the OSI model.
- It is easier to improve with advancements as each layer can get updates separately.

Layer No	Layer Name	Responsibility	Information Form(Data Unit)	Device or Protocol
7	Application Layer	Helps in identifying the client and synchronizing communication.	Message	SMTP
6	Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Message	JPEG, MPEG, GIF
5	Session Layer	Establishes Connection, Maintenance, Ensures Authentication and Ensures security.	Message	Gateway
4	Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segment	Firewall
3	Network Layer	Transmission of data from one host to another, located in different networks.	Packet	Router
2	Data Link Layer	Node to Node Delivery of Message.	Frame	Switch, Bridge
1	Physical Layer	Establishing Physical Connections between Devices.	Bits	Hub, Repeater, Modem, Cables



7 Layers of the OSI Model



7. Application

- End User layer
- HTTP, FTP, IRC, SSH, DNS



6. Presentation

- Syntax layer
- SSL, SSH, IMAP, FTP, MPEG, JPEG



5. Session

- Synch & send to port
- API's, Sockets, WinSock



4. Transport

- End-to-end connections
- TCP, UDP



3. Network

- Packets
- IP, ICMP, IPSec, IGMP



2. Data Link

- Frames
- Ethernet, PPP, Switch, Bridge



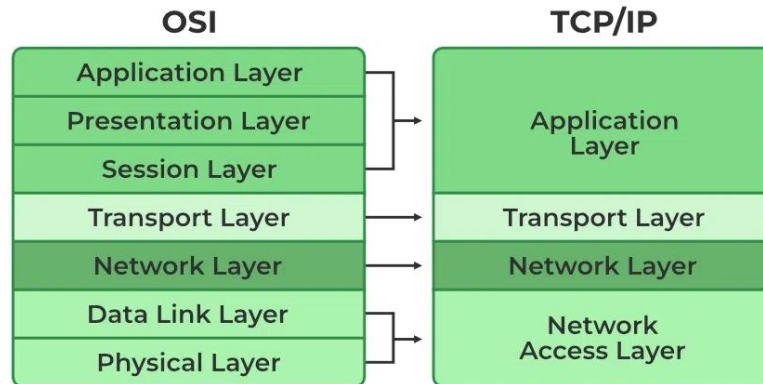
1. Physical

- Physical structure
- Coax, Fiber, Wireless, Hubs, Repeaters

OSI vs TCP/IP Model

Some key differences between the OSI model and the TCP/IP Model are:

1. TCP/IP model consists of 4 layers but OSI model has 7 layers. **Layers 5,6,7 of the OSI model are combined into the Application Layer of TCP/IP model and OSI layers 1 and 2 are combined into Network Access Layers of TCP/IP protocol.**
2. The TCP/IP model is older than the OSI model, hence it is a foundational protocol that defines how should data be transferred online.
3. Compared to the OSI model, the TCP/IP model has less strict layer boundaries.
4. All layers of the TCP/IP model are needed for data transmission but in the OSI model, some applications can skip certain layers. Only layers 1,2 and 3 of the OSI model are necessary for data transmission.



What Does TCP/IP Do?

The main work of TCP/IP is to transfer the data of a computer from one device to another. The main condition of this process is to make data reliable and accurate so that the receiver will receive the same information which is sent by the sender. To ensure that, each message reaches its final destination accurately, the TCP/IP model divides its data into packets and combines them at the other end, which helps in maintaining the accuracy of the data while transferring from one end to another end.

What is the Difference between TCP and IP?

TCP and IP are different protocols of Computer Networks. The basic difference between TCP (Transmission Control Protocol) and IP (Internet Protocol) is in the transmission of data. In simple words, IP finds the destination of the mail and TCP has the work to send and receive the mail. UDP is another protocol, which does not require IP to communicate with another computer. IP is required by only TCP. This is the basic difference between TCP and IP.

How Does the TCP/IP Model Work?

Whenever we want to send something over the internet using the TCP/IP Model, the TCP/IP Model divides the data into packets at the sender's end and the same packets have to be recombined at the receiver's end to form the same data, and this thing happens to maintain the accuracy of the data. TCP/IP model divides the data into a 4-layer procedure, where the data first go into this layer in one order and again in reverse order to get organized in the same way at the receiver's end.

1. Physical Layer: It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

2. Data Link Layer: The packet's network protocol type, in this case, TCP/IP, is identified by the data-link layer. Error prevention and "framing" are also provided by the data-link layer. Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

3. Internet Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows:

- **IP:** IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.
- **ICMP:** ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.
- **ARP:** ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP.

The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet. The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it.

Example: Imagine that you are using a computer to send an email to a friend. When you click “send,” the email is broken down into smaller packets of data, which are then sent to the Internet Layer for routing. The Internet Layer assigns an IP address to each packet and uses routing tables to determine the best route for the packet to take to reach its destination. The packet is then forwarded to the next hop on its route until it reaches its destination. When all of the packets have been delivered, your friend’s computer can reassemble them into the original email message.

In this example, the Internet Layer plays a crucial role in delivering the email from your computer to your friend’s computer. It uses IP addresses and routing tables to determine the best route for the packets to take, and it ensures that the packets are delivered to the correct destination. Without the Internet Layer, it would not be possible to send data across the Internet.

4. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

- **TCP:** Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

5. Application Layer

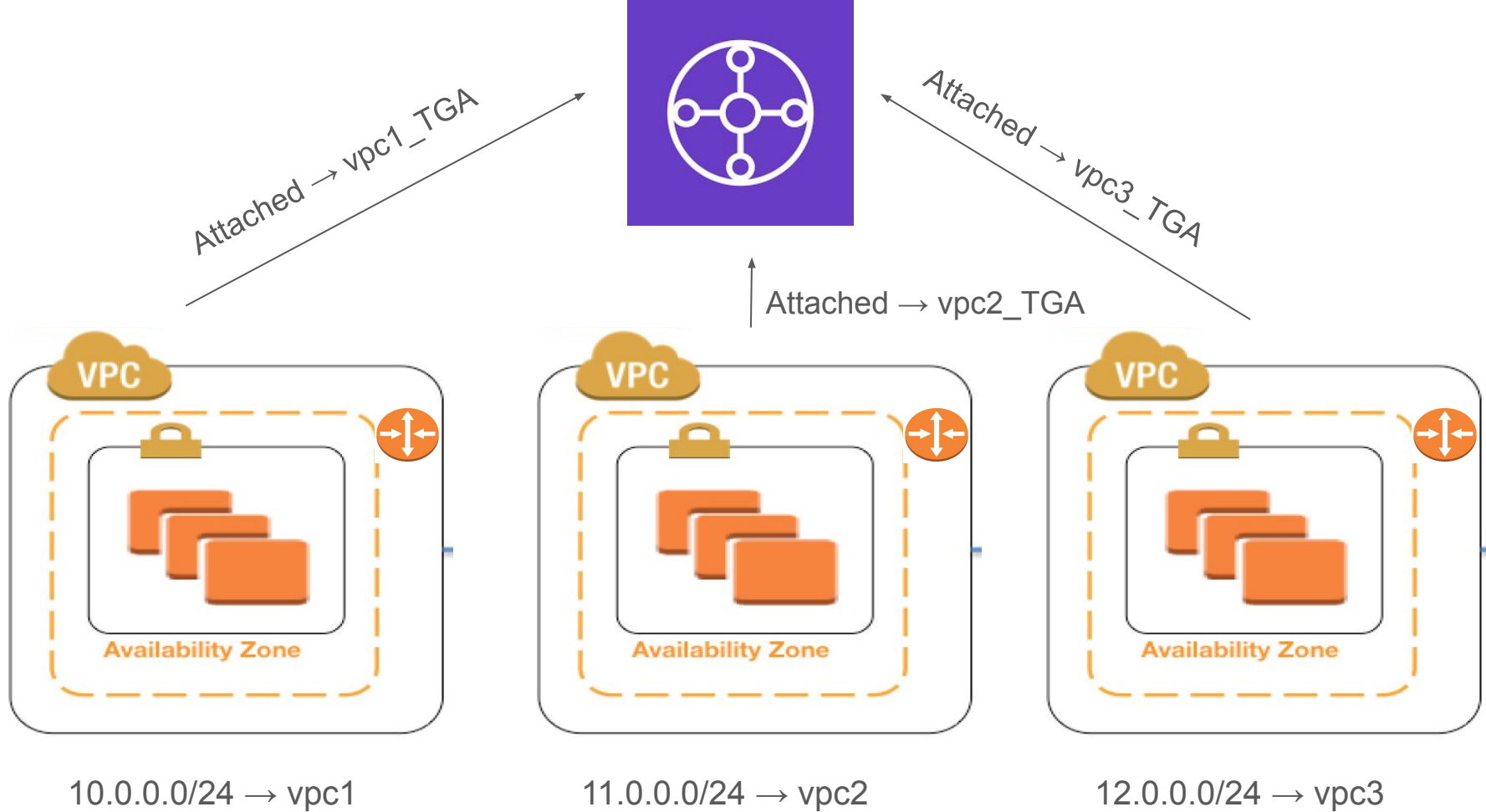
This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

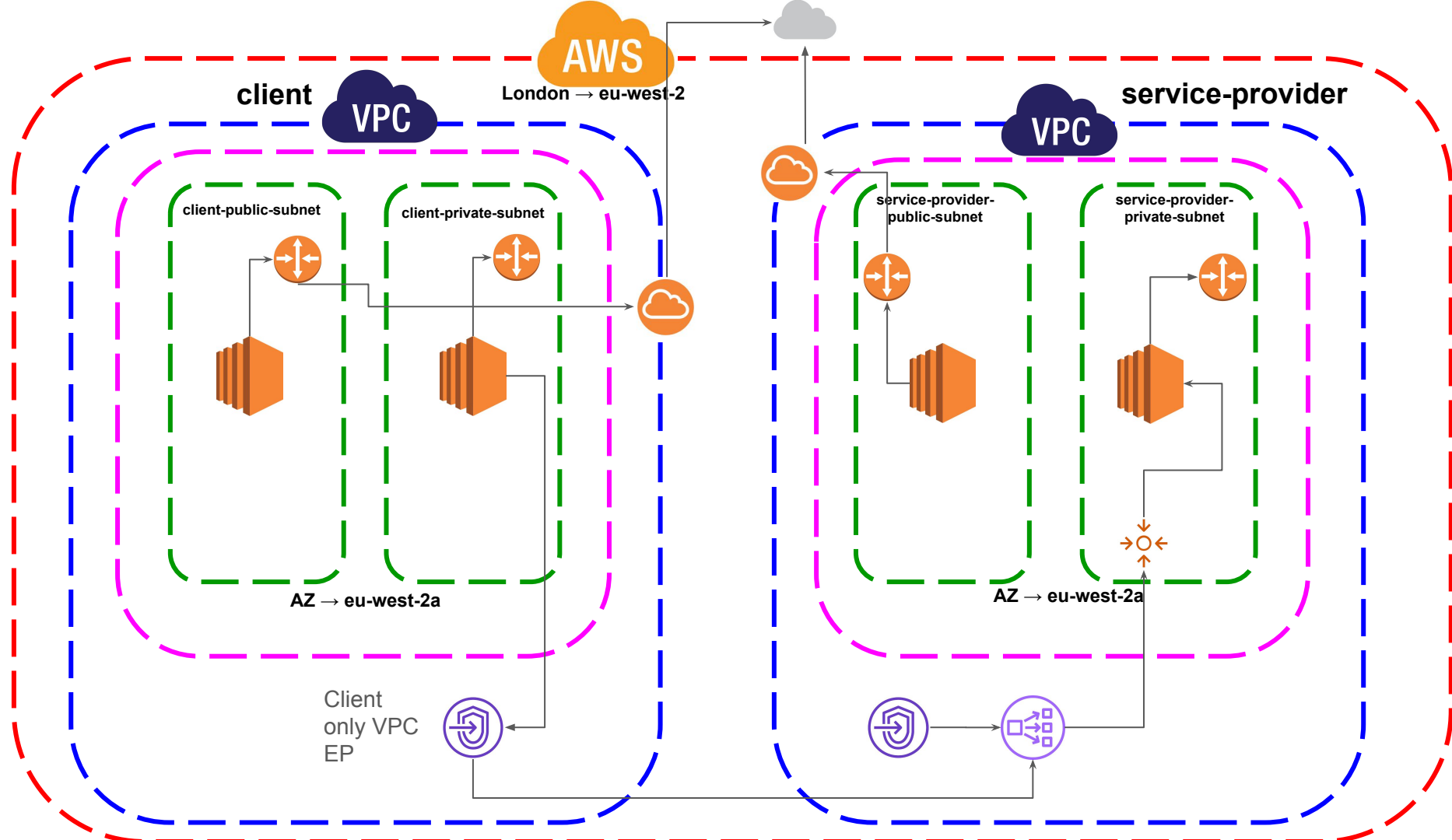
- **HTTP and HTTPS:** HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.
- **SSH:** SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:** NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

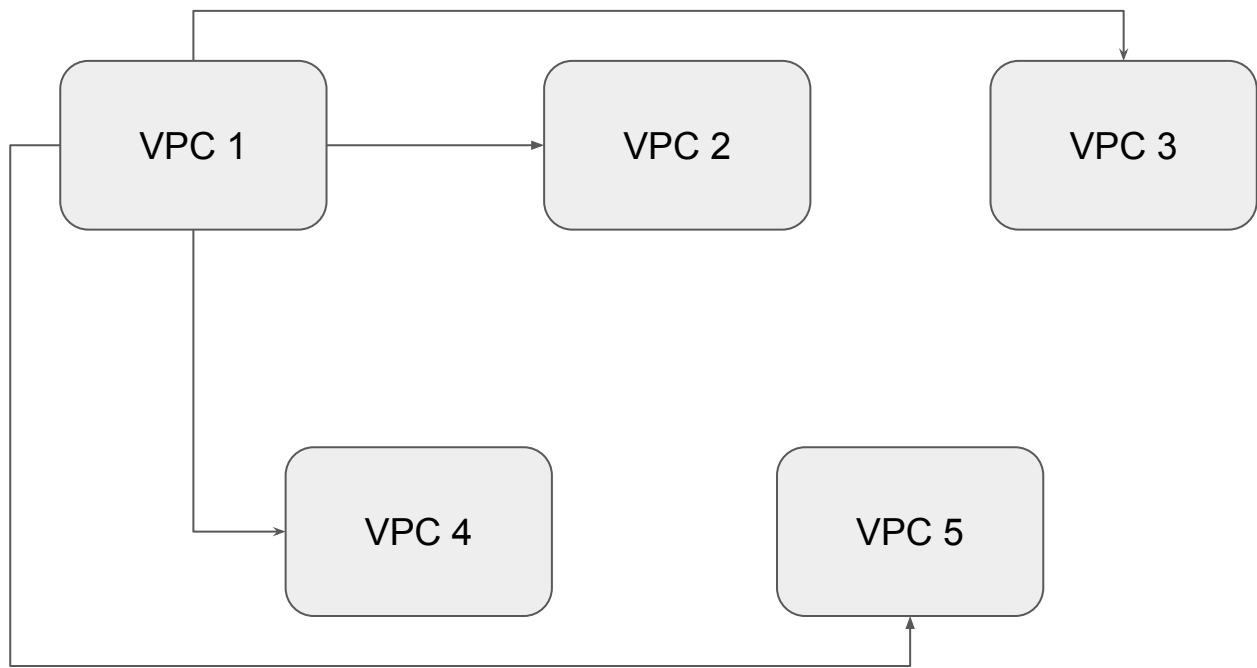
TCP/IP	OSI
TCP refers to Transmission Control Protocol.	OSI refers to Open Systems Interconnection.
TCP/IP uses both the session and presentation layer in the application layer itself.	OSI uses different session and presentation layers.
TCP/IP follows connectionless a horizontal approach.	OSI follows a vertical approach.
The Transport layer in TCP/IP does not provide assurance delivery of packets.	In the OSI model, the transport layer provides assurance delivery of packets.
Protocols cannot be replaced easily in TCP/IP model.	While in the OSI model, Protocols are better covered and are easy to replace with the technology change.
TCP/IP model network layer only provides connectionless (IP) services. The transport layer (TCP) provides connections.	Connectionless and connection-oriented services are provided by the network layer in the OSI model.

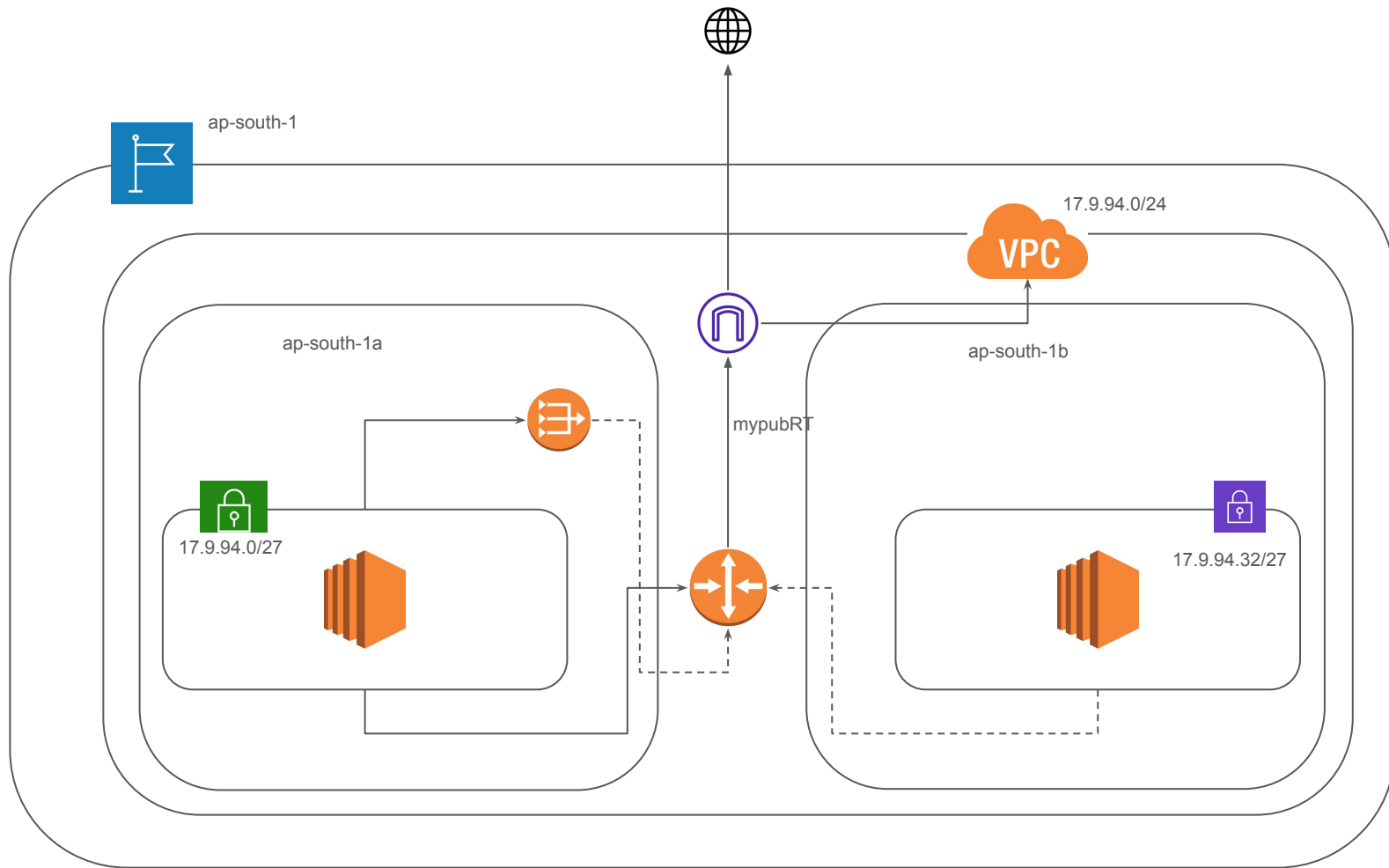
WAF understanding

- IP set → Define set of IPs that needs to be added as custom rule for WAF through WEB ACL
- Creation of WEB ACL
Step 2 and 3:
 - Custom rules: IP sets {Random IP numbers, or CIDRs of your organisation}
 - Action: ALLOW, BLOCK, CAPTCHA, CHALLENGE
 - Rule 1: Allow organization sites whitelisted in VPN (ALLOW)
 - Rule 2: Deny access to gaming sites, social media and entertainment sites, malicious sites, unverified sites whitelisted in VPN (BLOCK)
- If and else for step 3:
 - If the rules mentioned in rule 2 are okay, then ALLOW
 - If the rules mentioned in rule 2 are not okay, then BLOCK









VPC → 23.0.0.0/22 → 1024 IPs

Subnet 1: ALBpub1snet1a → 23.0.0.0/24

Subnet 2: ALBpub2snet1a → 23.0.2.0/25

Subnet 3: ALBpvtsnet1a → 23.0.2.128/25

Subnet 4: ALBpub1snet1b → 23.0.1.0/24

Subnet 5: ALBpub2snet1b → 23.0.3.0/25

Subnet 6: ALBpvtsnet1b → 23.0.3.128/25